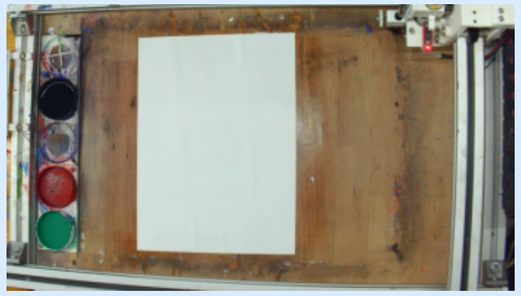
How the brush plotter paints a picture - overview (python paint.py ko/21.png --probe)

1 Camera + machine

home Z, then X Y; head parks out of view;
photo of the table. calibration.npz
(calibrate_dots.py: the brush paints 30
dots) + lens.json turn pixels into mm
-> paper_frame.png



2 Paper + placement

finds the sheet (any size, tilted ok)
turns the art 90 deg if it fits larger
centred in the margin; signature + year
in the bottom margin (painted first)
-> <name>_overlay.png



3 Colours + dishes

artwork -> one mask per colour,
black last (PAINT_COLOR 'black' =
all in black); each colour -> its
dish, found by hue in the photo
-> overlay circles



4 Probes

2 sloping lines, left + right edge
touch = where the line starts to widen
width at every depth, tilt, tip offset
preview opens, no question asked
-> probe_result.png



5 Planning

skeleton: every shape = centre-line
strokes, depth from width, side strokes
for wider parts; thin end first
(leaf / voronoi styles still there)
-> <name>_<colour>_strokes.png



6 Painting

glides in along the line, depth follows
width + tilt; dip after 70-90 mm,
wide strokes count more (by width)
Ctrl+C: brush up, then stop
-> out.gcode, paint0.png



7 Drying + repair

wait 20 min, photo vs start photo:
white spots + missing thin lines
painted again, a bit deeper each pass
until done or no progress
-> paint_final.png, paint_repairN.png



You decide

The probe plan picture opens by itself;
the probes are painted without asking
(PROBE_ASK = True: ask again).

Before the first art stroke: the overlay
and strokes pictures, then
'Start painting now? [y/N]'.

Probes unusable -> nothing is painted.
Ctrl+C at any time: brush up first.
All numbers: settings.py (page 7).

camera machine paints computer only

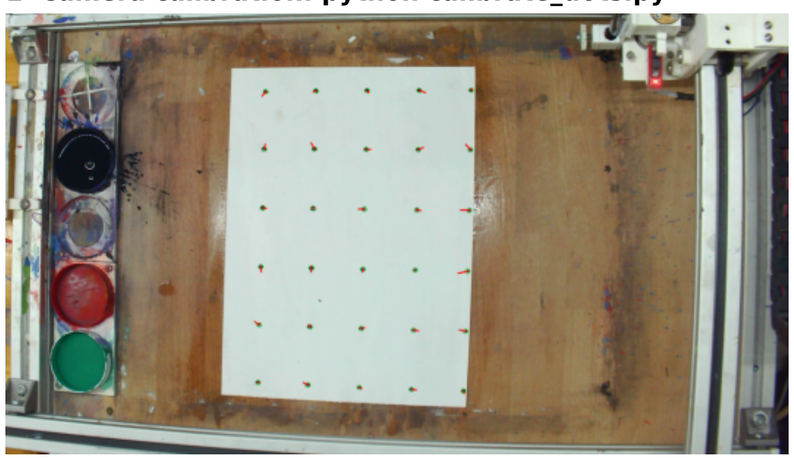
Chapters 1-3 - camera, calibration, paper, placement, signature, colours

1 The photo of the table



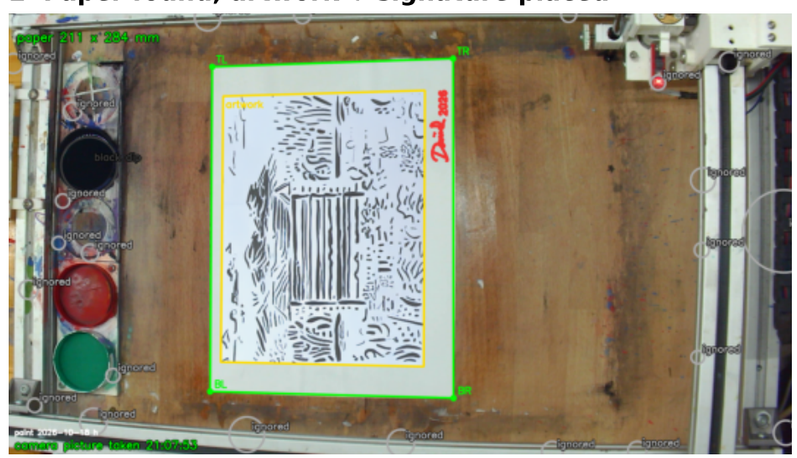
toolhead parked out of view; calibration.npz + lens.json turn every camera pixel into machine mm (and back)

1 Camera calibration: python calibrate_dots.py



the brush paints 30 dots at known machine positions, the camera finds them: 0.21 mm error (2026-11-02). Run it again when the camera moved (sign: probe marks beside the yellow lines, or off the paper)

2 Paper found, artwork + signature placed



green = the sheet, yellow = artwork box (ROTATE = auto: turned when it fits larger), red = signature + year (SIGNATURE, SIGN_WIDTH_MM) in the bottom margin; grey = ignored specks; 'black dip' = its dish

3 The artwork (ko/21.png)



split into colour layers by hue; dark pixels with little colour = black (DARK_CHROMA); light colours first, black last

3 Colours and dishes

```
one mask per colour; its dish = the paint
pool in the photo whose hue is nearest

PAINT_COLOR = 'black' -> everything in
  black, whatever colours the picture has
a dark weak colour layer -> merged into
  black (LAYER_MIN_CHROMA)
a tinted halo around black strokes
  -> black (LAYER_FRINGE_SHARE)
pools: >= 300 mm2, round enough, within
  200 mm of DIP_X/DIP_Y

every layer planned + painted on its own;
rinse in the water dish in between
(WATER_X / WATER_Y, if set)
```

2 Signature and margins

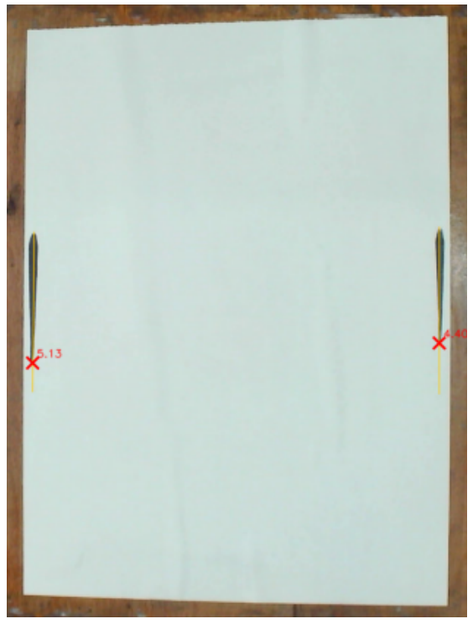
```
SIGNATURE   podpis2.png
  35 mm wide, year 6 mm high
  4 mm under the artwork, 6 mm
  paper left free below it
  lines 1.6 mm wide (depth from the
  probe curve)
  painted FIRST, right after the probes

the reach of the machine limits a big
sheet (MACHINE_X/Y_MIN/MAX); margins
stay even on both sides

the probe lines take the outer 5 mm
(PROBE_INSET_MM) of the left + right
edge; a second probe on the same sheet
goes beside the first
```

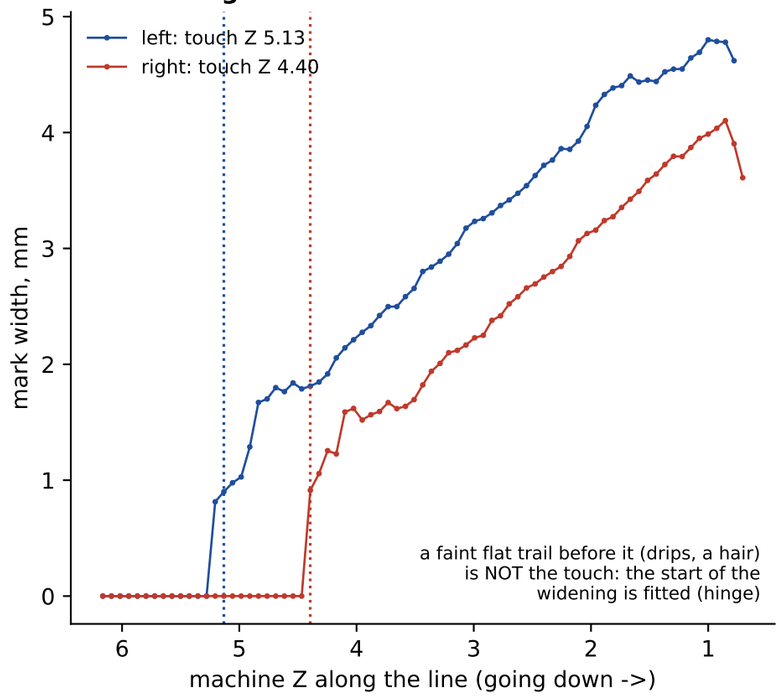
Chapter 4 - probes: where THIS brush touches THIS paper, and how wide it paints

Two sloping lines (left + right edge)

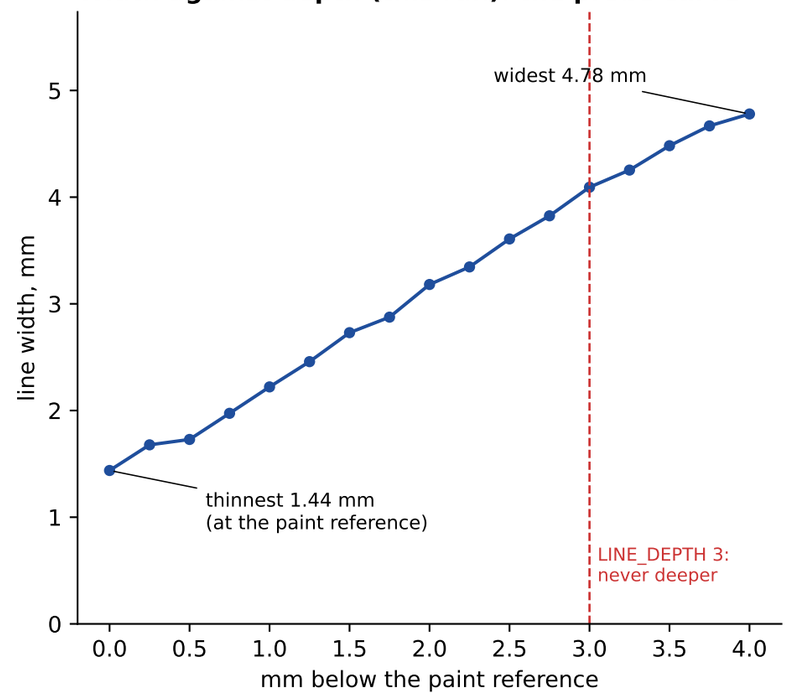


each line goes down from 2 mm above the expected touch to 4 mm below it (one dip). Yellow = where the camera measures, red x = the touch, with its machine Z

Width along each line: touch = where it starts to widen

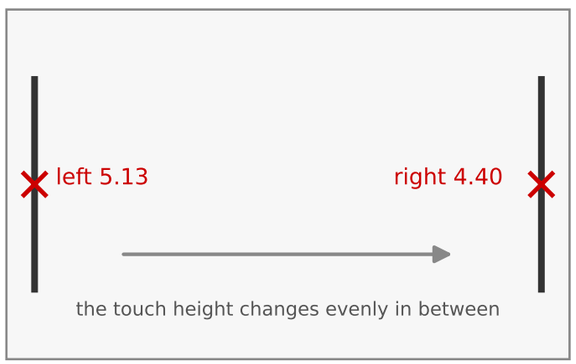


Width against depth (this run): the probe curve



Paper tilt: a straight slope from left to right

PROBE_COUNT = 2: left + right edge, where the paper lies flattest. (4 = one line per edge: a plane, cross-checked)



Where the paint really lands

tip offset now (+0.00, -0.14) mm: measured ACROSS the two lines from the photo (along them: the last probe's value); every X Y of the job is shifted by the opposite.

Next probe: its lines are already moved by the last offset, so a steady offset is used up front.

A probe mark beside its yellow line by many mm, both lines the same way = the camera moved: run calibrate_dots.py (2026-11-02: 9.5 mm, right line on the wood). Larger than 10 mm (PROBE_OFFSET_MAX_MM): not used.

What the probes set for this job

```
touch (where it widens)  Z 4.75
paint reference          Z 4.50
  = touch - 0.25 (PROBE_MARGIN_MM)
thinnest line            1.44 mm
widest mark              4.78 mm
tilt left -> right       +0.74 mm
tip offset               (+0.00, -0.14) mm

Z of every stroke = reference
  - depth(width) + tilt there
the repair uses the same values

NOT used -> the job stops, nothing
painted: no paint seen, never wide
enough, marks already at the start
(--unmeasured paints anyway)
```

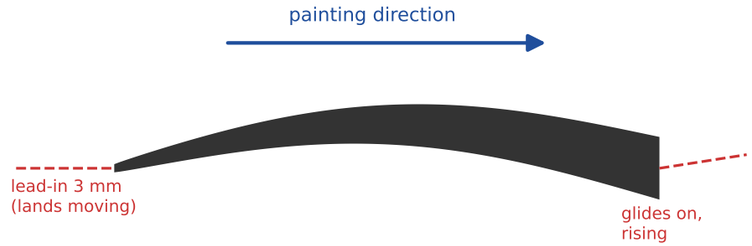
Chapter 5 - planning: every shape as centre-line strokes (INFILL_STYLE = 'skeleton')

The strokes picture (ko/21, black, made before painting)



left: every stroke in its own colour, along the centre line of its shape; right: the paint at the real width. Title line: strokes + real coverage

One line: thin end first, glide out at the thick end



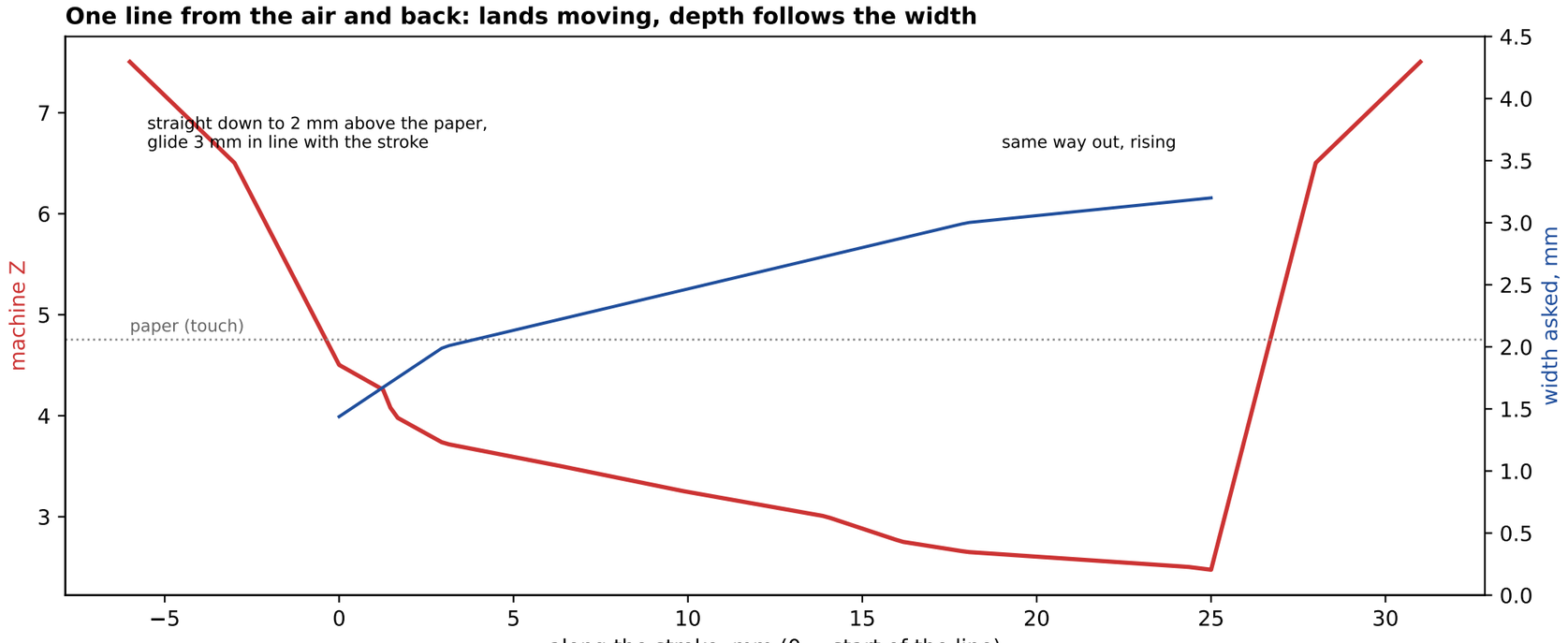
the end with the most extra width (centre of the width along the whole line) is painted LAST; depth follows the width; pointed ends: run on to 0.5 mm wide (LINE_TIP_MM)

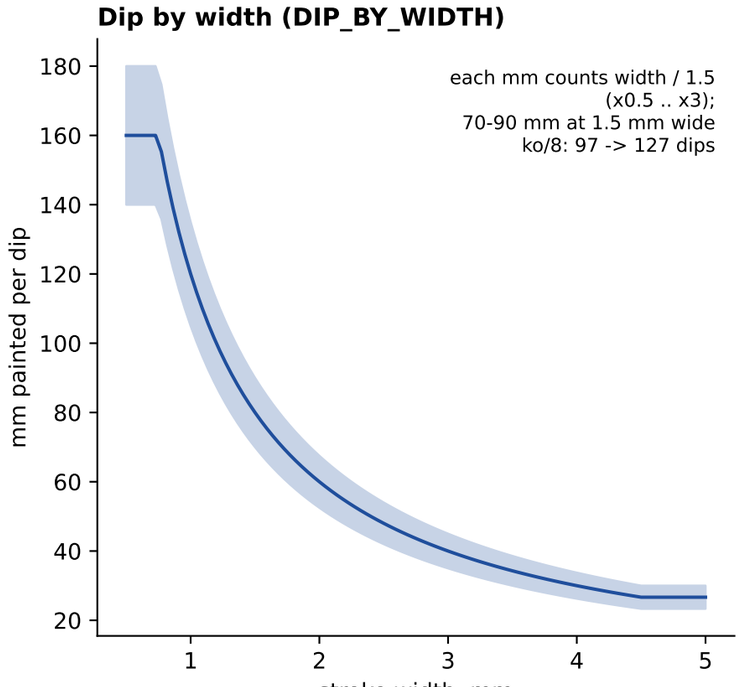
How shapes become strokes

```
every shape: one stroke along its centre
  line, pressed deeper where it is wider
up to 1.5 x the widest mark: still ONE
  stroke (LINE_WIDTH_SLACK)
wider: what is still open gets side
  strokes along ITS centre line
  (up to 4 rounds)
next stroke = the nearest (short travel)

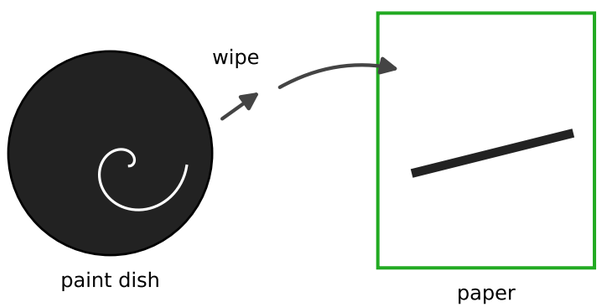
other styles (INFILL_STYLE, or per
colour INFILL_STYLE_BY_COLOR):
  leaf    = Voronoi leaves made of veins
  voronoi = calligraphic dashes
  cells, bands, wide, lines = older
REGION_STYLES: long thin pieces in
  another style
```

Chapter 6 - painting: landing, depth, dips





The dip cycle



down into the dish (2 mm above touch), a spiral,
wipe on the rim towards the next stroke, back to the
exact point where the stroke stopped (lead-in again);
dip point wanders 10 mm (not one spot used up)

What every move gets

```
X Y  machine mm from the placement
     - tip offset (lands where planned)
Z    paint reference - depth(width)
     + tilt there; never deeper than
     LINE_DEPTH 3 mm
F    paint 3000 mm/min, accel 200
     travel 12000, lifted 3 mm
     to a dish 15 mm high

a line that does not fit in the paint
left: dip BEFORE it (no dip mid-line
unless it is longer than one dip)
```

During the job

```
PHOTO_EVERY_DIPS = 0
  0 = live window, start + final photo
  5 = photo + overlay every 5 dips
STROKE_PHOTOS = 'off'
  'stroke' / 'leaf': a photo after each
  -> strokes_<time>.mp4
TIMELAPSE = False: camera video of the job
AUTO_GIF = False

colour finished -> rinse -> next colour
Ctrl+C: brush up, then stop
do NOT edit the .py files while a job
runs (the repair re-loads them)
```

Chapter 7 - drying, final photo, repair of white spots and missing lines

1 Final photo: what is still empty (ko/19)



after DRY_WAIT_MIN = 20 (wet gouache shines). magenta = the art as planned, yellow = still white inside it. R / Enter = photo now

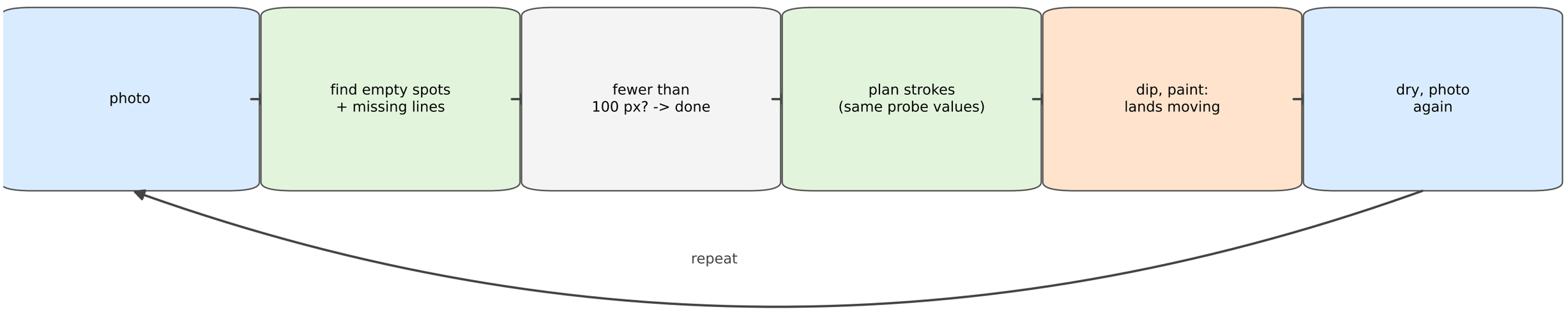
2 After the repair pass



only the empty spots got paint; edges are not painted over

How a spot counts as empty

```
the art has colour here AND the photo,
compared with the START photo, got no
darker (JUDGE_BY_DIFFERENCE)
minus strips along outlines thinner
  than 3 mm, within 0.5 mm of the edge
minus specks under 2 mm2; pinholes
  closer than 1.5 mm merge
thin lines (< 2.5 mm): checked along
  their centre line, repainted where no
  paint is within 0.7 mm, over >= 2 mm
paint > 1.5 mm from the plan: no repair
  (the photo would mislead it)
```



Repair strokes: pressed for at most 90 % of the widest mark (REPAIR_WIDTH_SHARE), nothing shorter than 1.2 mm. Each pass 0.15 mm deeper (at most 1).
Stops when the spots are gone, or a pass keeps more than 1.25 x the empty pixels of the one before. Dips in the repair: by length (unchanged).

Files a job writes, and the settings you will touch most

Files (in the program folder; <name> = next to the artwork)

paper_frame.png
the photo the paper was found in

probe_plan_<time>.png / probe_result.png
where the lines go (opens) / what was measured

probe_result.json
touch, curve, tilt, tip offset (reused next time)

<name>.gcode
the job (header: paper, probes, offset)

<name>_overlay.png
plan on the photo: paper, box, signature, dishes

<name>_<colour>_strokes.png
every stroke, and the paint at real width

<name>_<colour>_preview.png
mask vs plan: missed (red) / spill (blue)

paint0.png
start photo (all checks compare with it)

paint_final.png + _overlay
after drying

gap_paint_*_empty.png
the empty spots the repair paints

paint_repairN.png
after every repair pass

calib_dots_result.png
camera calibration check

crash_log.txt
errors (a crash is written here)

Settings (settings.py)

TOUCH_Z
rough touch height (4.2); the probe measures the real one

PROBES / PROBE_COUNT / PROBE_ASK
probe every job (True); 2 lines; ask first (False)

PROBE_ABOVE_MM / PROBE_BELOW_MM
line from 2 above to 4 below the touch

PROBE_MARGIN_MM / PROBE_Z_EXTRA_MM
paint a bit deeper than the touch / own correction

LINE_DEPTH
never deeper than this (3 mm)

LINE_LEAD_IN_MM / LINE_LAND_HIGH_MM
landing: glide length / height it starts from

INFILL_STYLE (+ _BY_COLOR)
'skeleton' | leaf | voronoi | cells | lines ...

DIP_MIN_MM / DIP_MAX_MM
paint between dips (70-90 mm)

DIP_BY_WIDTH / DIP_REF_WIDTH_MM
wide strokes dip sooner (True, 1.5 mm)

SIGNATURE / SIGN_*
signature file, size, year, place

PAINT_COLOR
'auto' = every colour own dish; 'black' = all black

DRY_WAIT_MIN / REPAIR_AT_END
wait before the final photo (20); repair on

REPAIR_*
spot size, rim, thin lines, depth steps